BANK OF PAPUA NEW GUINEA

Presentation on BPNG's Nowcasting Process

Presentation by PNG team | 30th September 2026

IMF Cohort Training in Nowcasting and forecasting- Part II | STI- Singapore

Contents of Presentation

1 FPAS Progressive Update- Overview

2 BPNG's nowcasting Process

3 Nowcasting Real GDP

4 Nowcasting CPI Inflation

5 Challenges and Way Forward

6 Conclusion

1. FPAS Update- Overview

- About Seven (7) TA missions so far (2024-2026) on FPAS/Nowcasting
- TA Missions focused on developing and enhancing BPNG's FPAS- Development of Nowcasting tools/models, staff capacity building, communication, and entire forecasting and policy analysis system- (Key element of FPAS.
- Tools:
  - Nowcasting models for Total Real GDP and GDP by sector
  - Nowcasting database- (excel based)
  - Macro model- MFF (Oil Price shock scenario, El Nino scenario)

- People
  - Staff continue to receive training and support in terms of nowcasting/forecasting/modelling and data analysis

- Communication
  - GDP and inflation nowcasting results communicated and shared with MPC, Board, senior management, etc.
  - Nowcasting results help inform and guide policy discussions, especially during the drafting of the March and September Economic Outlook.

- System- MEPG restructure- approval given for establishment of modelling unit and statistics department.

2. Real GDP Nowcast

2. BPNG's Nowcasting Process

1. Database Update

- Nowcasting database is updated on a quarterly basis
- Data sourced from BPNG QEB tables and various official sources

2. Data Processing

- Data is imported into Eviews and processed (e.g. Data transformation, seasonal adjustment, frequency conversion, etc)

3. Estimate/Run Bridge Equations

- Nowcast total Real GDP
- Employ machine learning techniques for selecting independent variables
- Model selection criteria: R-squared, AIC, RMSE, DW stat

4. Discuss and communicate model results

- GDP modelling team discuss the results.
- These results guide policy discussions during economic projections and MPC meetings
- Nowcasting report included in MPC information papers.

2. Nowcasting Real GDP

We approach GDP Nowcasting in two ways:

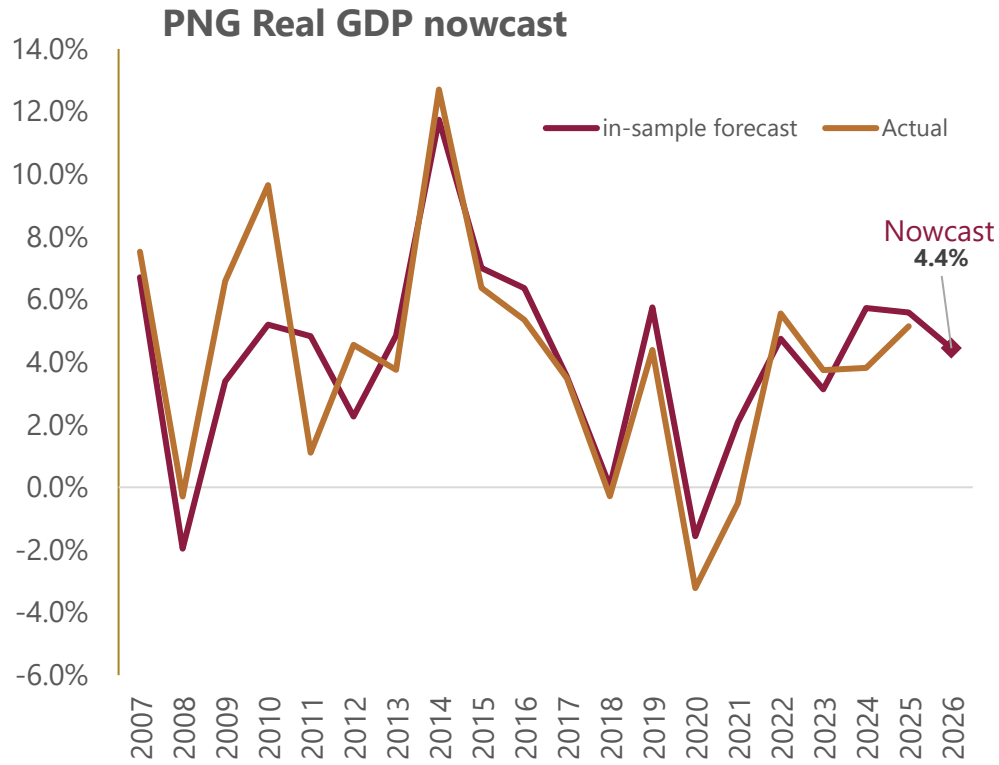
1. **Nowcast Total Real GDP**
    - We have developed several models which we use to nowcast total real GDP.

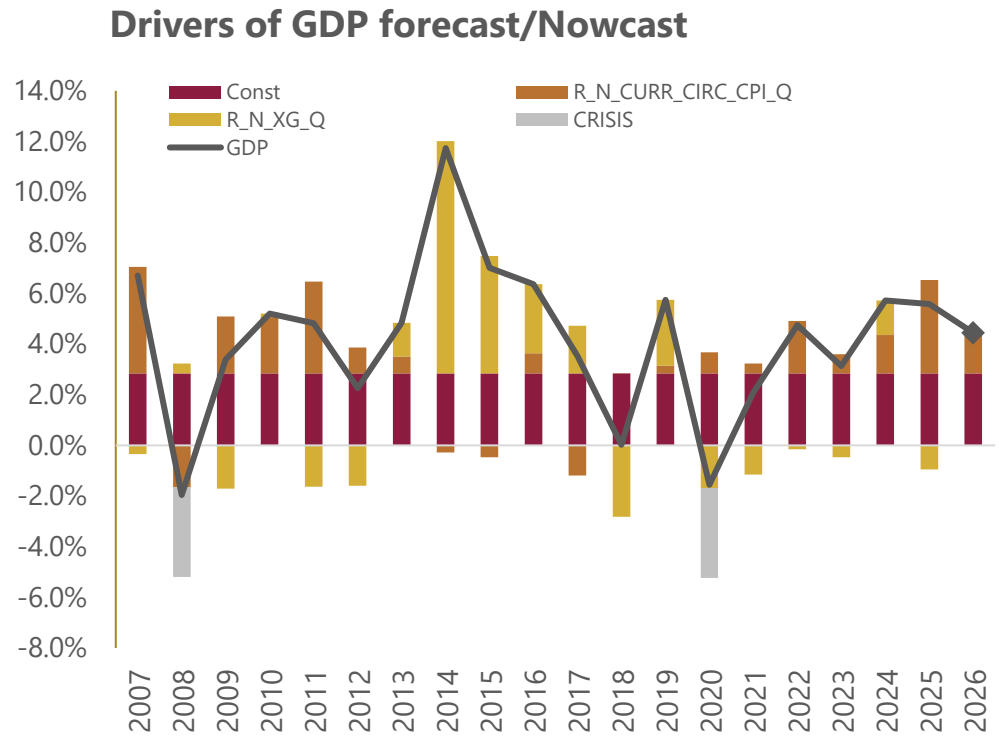
2. Nowcast Real Gross Value Added for the different industries:

- These sectors are Agriculture Forestry & Fisheries (AFF), Mining and Quarrying, Construction, services, etc.
- Some industries are quite difficult to nowcast, hence we use average growth rates instead.
- Total Real GDP= sum of industry gross value added- taxes less subsidies.

Different models give slightly different results. Therefore, we get the average results of all models to arrive at our final nowcast of real GDP.

2. Nowcasting Real GDP (cont.)





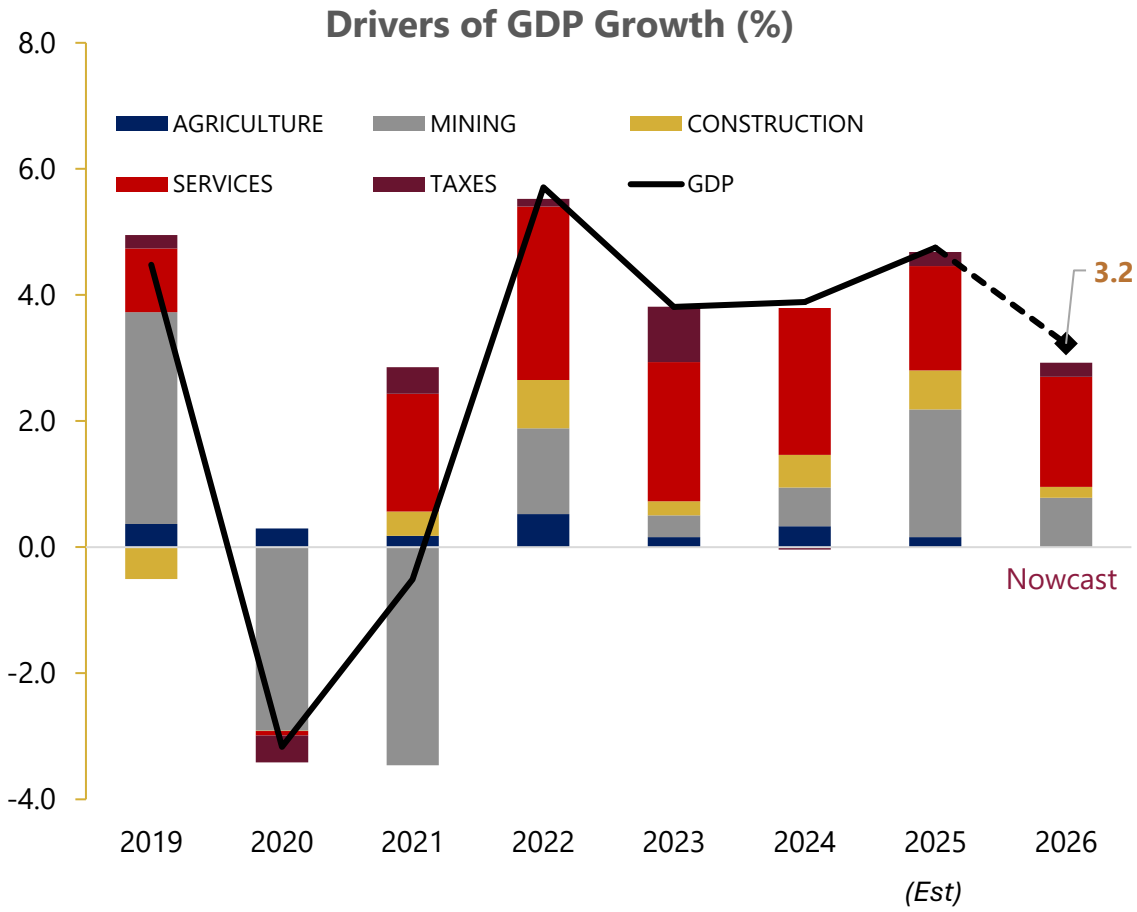
HF indicators used:

- Currency in circulation
- Total exports
- Crisis dummy (covid and GFC)

Note: 2026 GDP nowcast updated using Q1 Data

2. Nowcasting Real GDP by Industry GVA

	TABLE 9. INDUSTRY GROSS VALUE ADDED, CONSTANT PRICE ESTIMATES - PERCENTAGE (%) CHANGE (a)	2021	2022	2023	2024	2025	2026
A	AGRICULTURE, FORESTRY AND FISHING	1.1%	3.1%	1.0%	2.1%	1.0%	1.5%
B	MINING AND QUARRYING	-11.6%	5.1%	1.3%	2.4%	8.0%	3.0%
C	MANUFACTURING	7.3%	10.4%	-0.9%	6.3%	2.7%	3.2%
D	ELECTRICITY,GAS,STEAM AND AIRCONDITIONING SU	-2.5%	0.5%	1.1%	2.6%	1.7%	1.3%
E	WATER SUPPLY, SEWAGE, WASTE MANAGEMENT	6.7%	7.3%	0.0%	4.8%	4.9%	5.0%
F	CONSTRUCTION	7.2%	13.4%	3.7%	8.4%	9.7%	2.6%
G	WHOLESALE AND RETAIL TRADE	5.9%	6.8%	2.8%	2.9%	3.1%	3.4%
H	TRANSPORT AND STORAGE	3.0%	12.2%	5.8%	5.4%	1.6%	1.3%
I	ACCOMODATION AND FOOD SERVICE ACTIVITY	4.7%	6.2%	3.2%	3.6%	1.3%	1.8%
J	INFORMATION AND COMMUNICATION	18.8%	12.6%	15.5%	15.6%	8.0%	9.0%
K	FINANCIAL AND INSURANCE ACTIVITIES	-0.4%	3.5%	4.7%	5.4%	-0.4%	0.3%
L	REAL ESTATE ACTIVITIES	0.3%	2.8%	3.0%	3.3%	2.5%	2.4%
M	PROFESSIONAL,SCIENTIFIC AND TECHNICAL ACTIVIT	-1.8%	4.6%	3.8%	3.9%	3.7%	2.4%
N	ADMINISTRATIVE AND SUPPORT SERVICE ACTIVITIES	0.7%	9.1%	2.7%	4.3%	3.1%	2.3%
O	PUBLIC ADMINISTRATION AND DEFENCE, COMPULSC	7.5%	3.8%	9.7%	3.3%	5.6%	6.5%
P	EDUCATION	6.0%	3.4%	5.5%	14.0%	5.1%	5.9%
Q	HUMAN HEALTH AND SOCIAL WORK ACTIVITIES	8.3%	3.3%	2.5%	2.0%	4.4%	5.1%
R	ARTS,ENTERTAINMENTAND RECREATION	7.2%	5.9%	2.3%	1.7%	2.2%	2.5%
S	OTHER SERVICE ACTIVITIES	-2.3%	13.9%	8.0%	9.5%	2.9%	2.7%
T	ACTIVITIES OF HOUSEHOLDS AS EMPLOYERS	3.1%	3.1%	3.1%	3.1%	3.1%	3.1%
	GVA at Basic Prices	**-1.0%**	**5.8%**	**3.1%**	**4.1%**	**4.8%**	**3.2%**
	Taxes less subsidies on products	10.5%	2.7%	20.4%	-0.7%	4.7%	4.6%



Bridge models used to nowcast RGVA for AFF, Mining, Construction, Wholesale and retail, ICT, Admin and support services.

2. Nowcasting Real GDP

HF variable selection for bridge models- Python program

Python_PNG_new.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

[61] ✓ 0s

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import IPython, graphviz, re, pydotplus, pydot
from sklearn.preprocessing import normalize, StandardScaler
from sklearn.neighbors import KNeighborsRegressor
from sklearn.tree import DecisionTreeRegressor
from sklearn.ensemble import RandomForestRegressor,GradientBoostingRegressor
from sklearn.model_selection import GridSearchCV, train_test_split, RandomizedSearchCV
from sklearn.metrics import mean_squared_error as MSE
from scipy.cluster.hierarchy import linkage, dendrogram
from pprint import pprint
from sklearn.tree import export_graphviz
import abc, collections, hashlib, itertools, json, operator, pathlib
import mimetypes, inspect, typing, functools, importlib, weakref
from sklearn import tree
```

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```
#CHOOSE THE PREDICTOR FROM THE DATABASE
Predictor = 'r_gva_min'
```

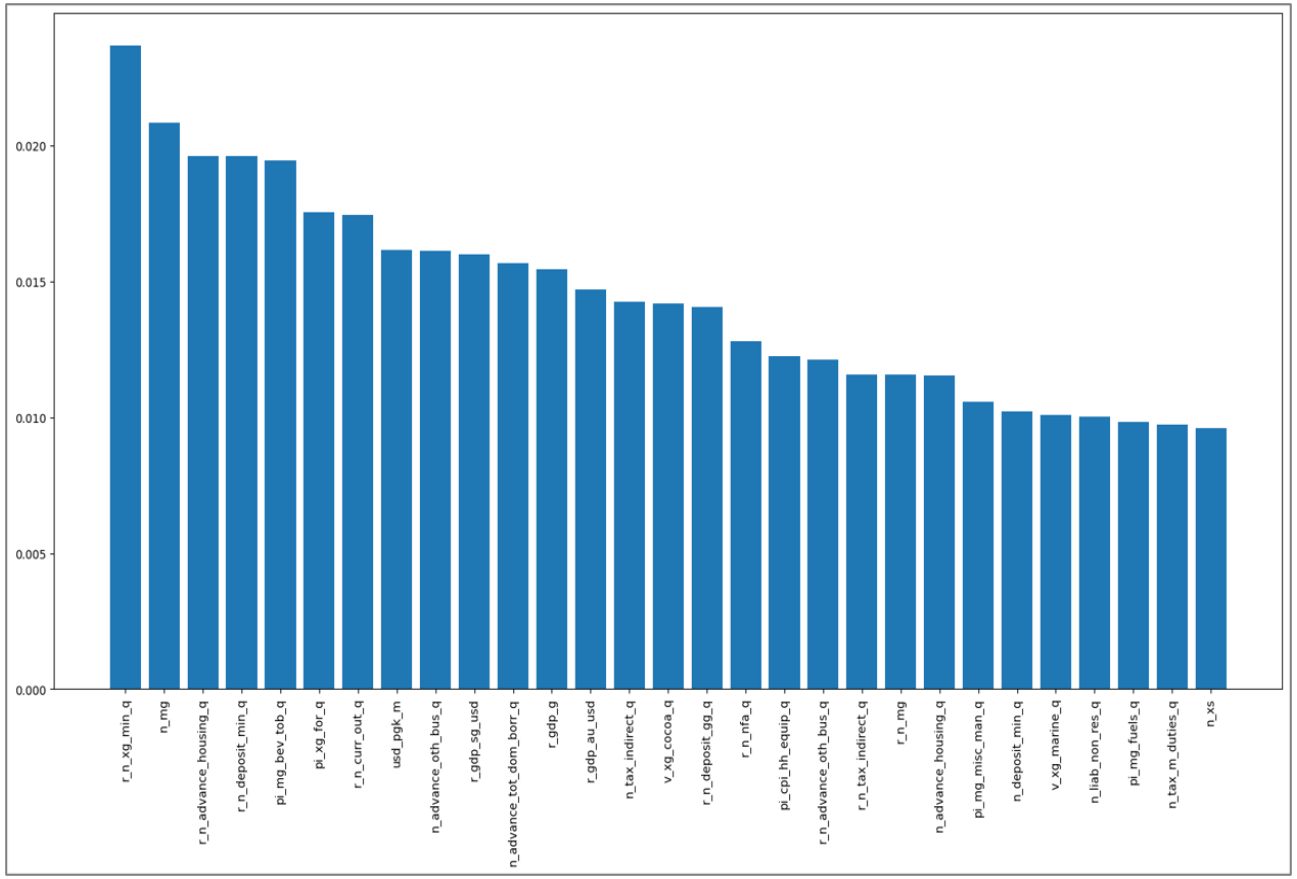
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```
# Some functions for using later

# Sort Features ascending in the DataFrame
def rf_feat_importance(m, df):
    return pd.DataFrame({'cols':df.columns, 'imp':m.feature_importances_}
                        ).sort_values('imp', ascending=False)
```

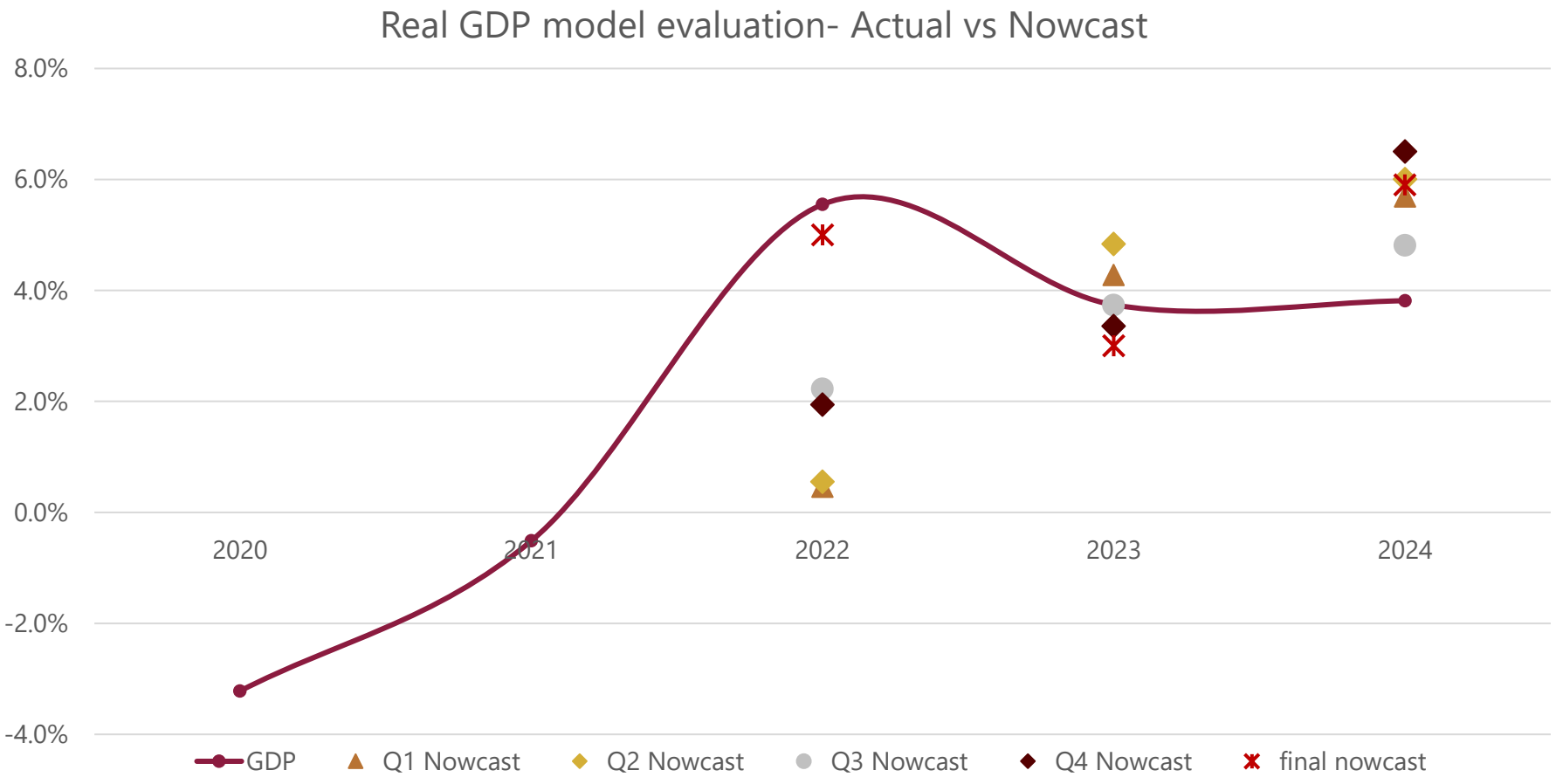
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```
# Read the csv file and display the data
df = pd.read_excel(r'nwc_trans.xlsx')
```



3. Nowcasting Real GDP

Out of sample model testing- Example (2022-2024)



Note: Final nowcast uses updated actual data for HF indicators

3. Nowcasting Headline Inflation

Systematic Selection of High Frequency Indicators Using Python

1. Indicator Screening



Test each possible indicator individually

* Run regressions of CPI inflation on each indicator

* Test contemporaneous and lagged relationship

* Select the top performing indicators based on:
  - coefficient and signs
  - p-value
  - R^2

2. Build Candidate Models



Combine selected indicators into regression models

* Test different combinations of indicators (lags)

* Filter for best models based on:
  - Adjusted R^2
  - AIC

3. Out of Sample Evaluation



Assess Forecasting Performance

* Fit data into training and testing periods

* Estimate model using training sample

* Generate forecast for test periods

* Evaluate forecast accuracy using RMSE

4. Final Selection



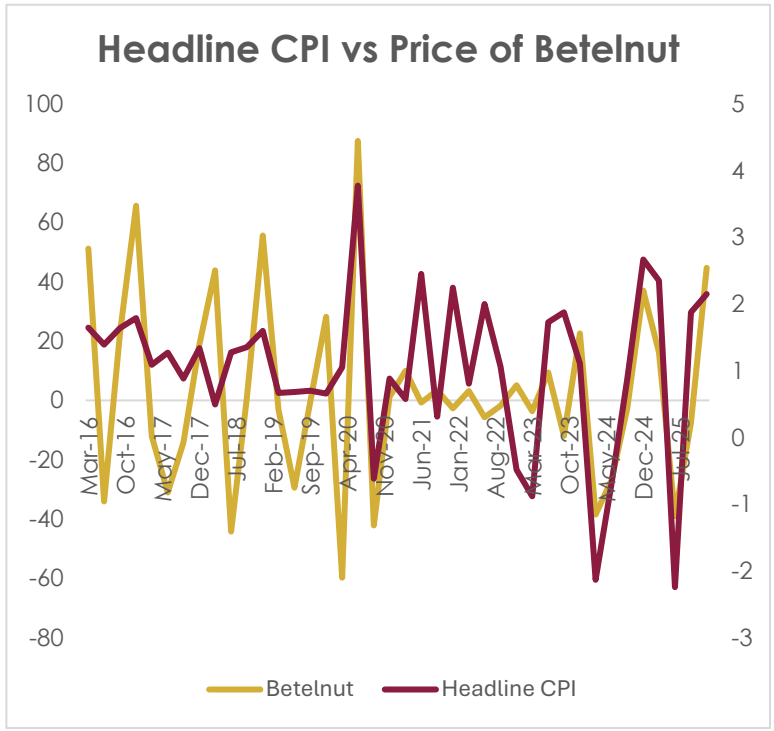
Select indicators and models based on forecasting performance

* Consider both statistical results and economic reasoning

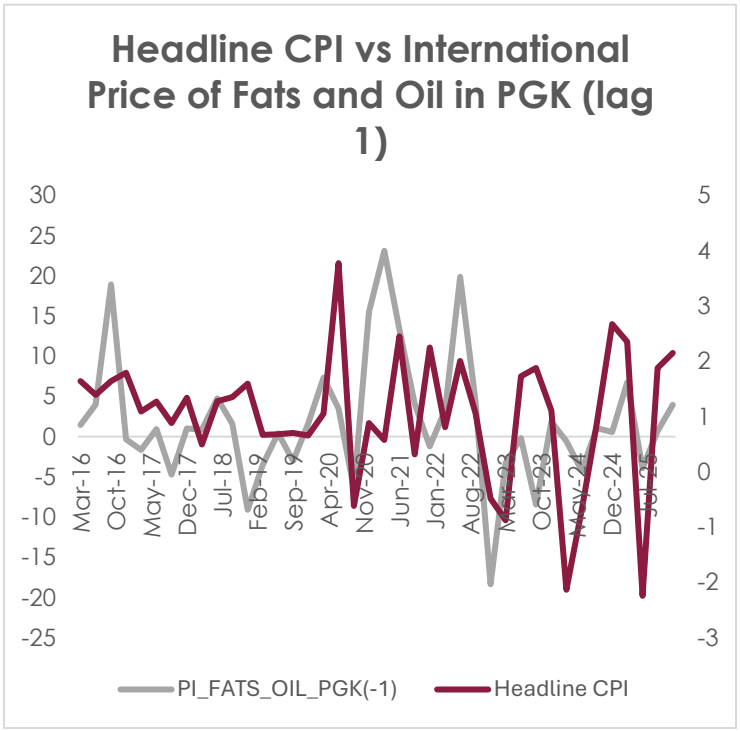
* final set of indicators are used in inflation nowcasting

Method 1: Nowcasting Quarterly Headline Inflation

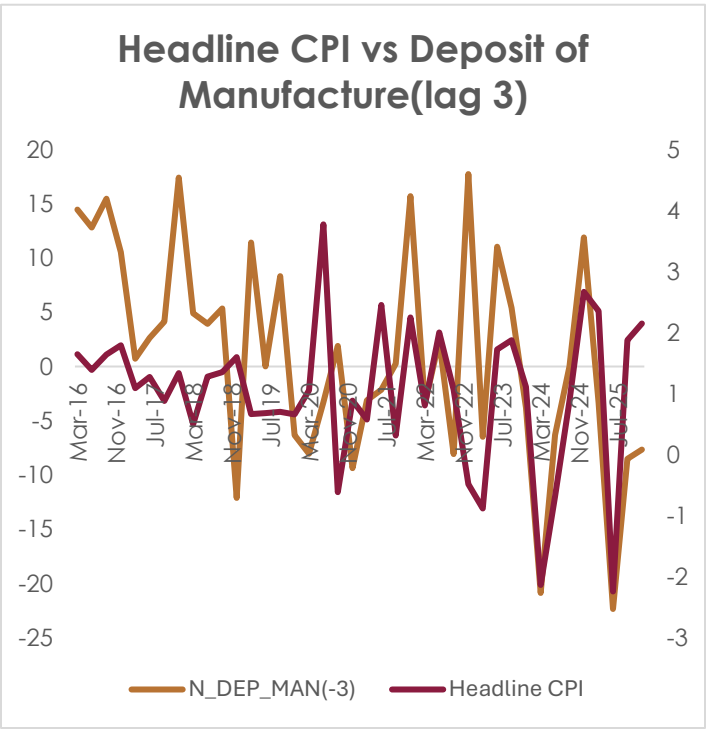
Leading Indicators of Headline Inflation



Domestic supply shocks (price shocks)



Imported Inflation



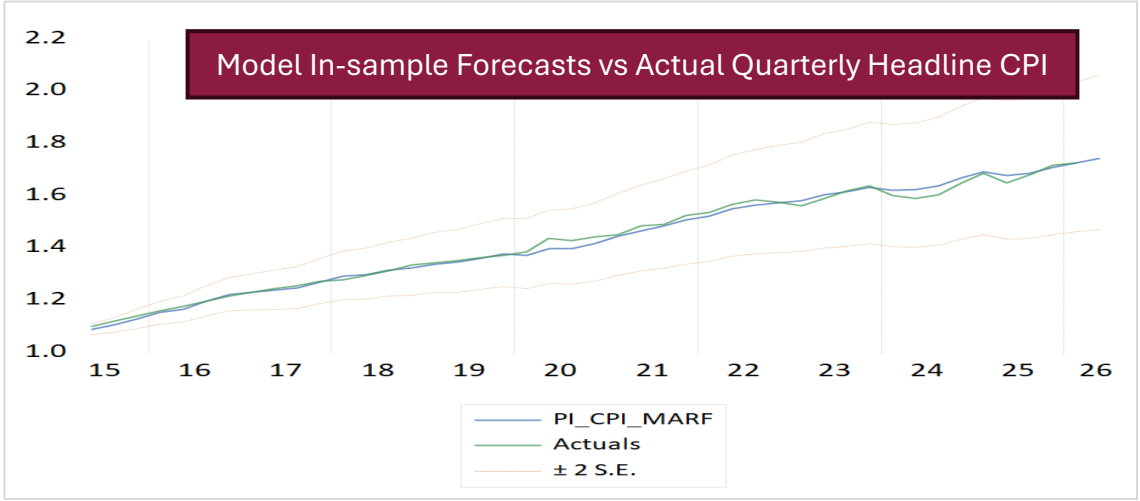
Domestic demand/liquidity conditions

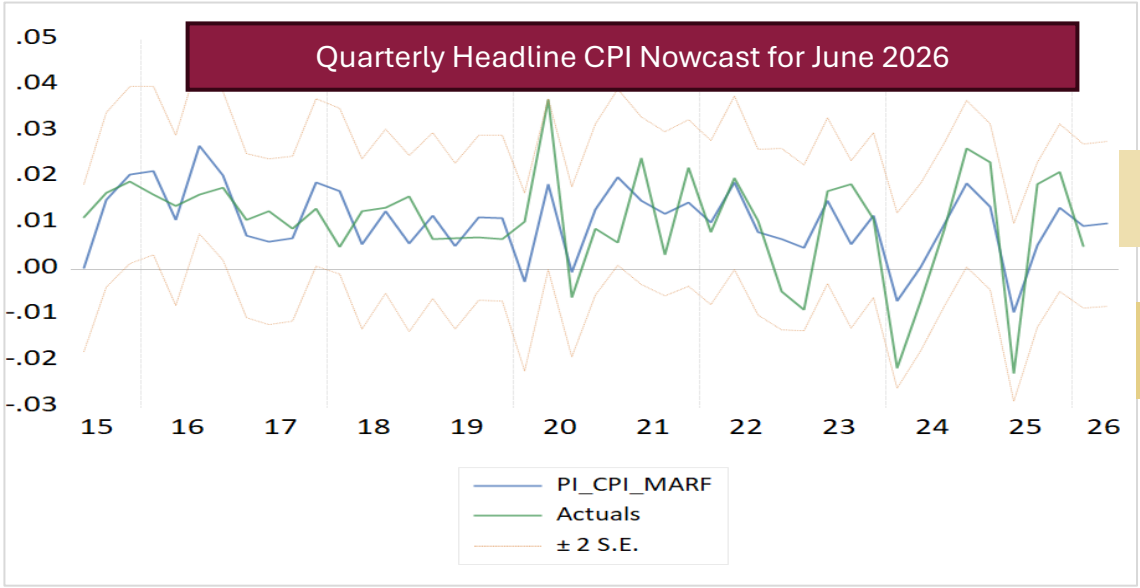
Method 1: Nowcasting Quarterly Headline Inflation

Dependent Variable: DLOG(PI_CPI_MAR26)
Method: Least Squares
Date: 05/25/26 Time: 09:53
Sample (adjusted): 2015Q2 2026Q1
Included observations: 44 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.009592	0.001358	7.062406	0.0000
DLOG(RPI_BETELNUT_F)	0.013737	0.004001	3.433323	0.0014
DLOG(N_DEPOSIT_MAN_F(-3))	0.040338	0.013254	3.043372	0.0041
DLOG(PI_FATS_OILS_PGK_F(-1))	0.050484	0.018493	2.729880	0.0094

R-squared	0.450434	Mean dependent var	0.010443
Adjusted R-squared	0.409217	S.D. dependent var	0.011460
S.E. of regression	0.008809	Akaike info criterion	-6.539691
Sum squared resid	0.003104	Schwarz criterion	-6.377492
Log likelihood	147.8732	Hannan-Quinn criter.	-6.479539
F-statistic	10.92826	Durbin-Watson stat	2.019259
Prob(F-statistic)	0.000022		



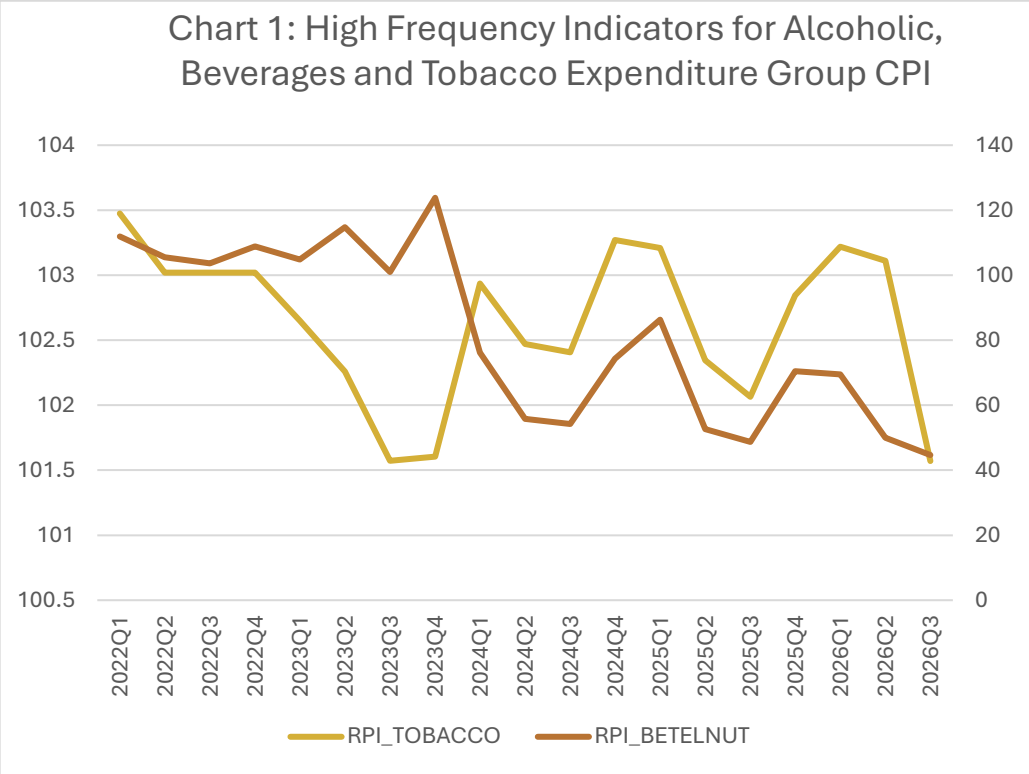


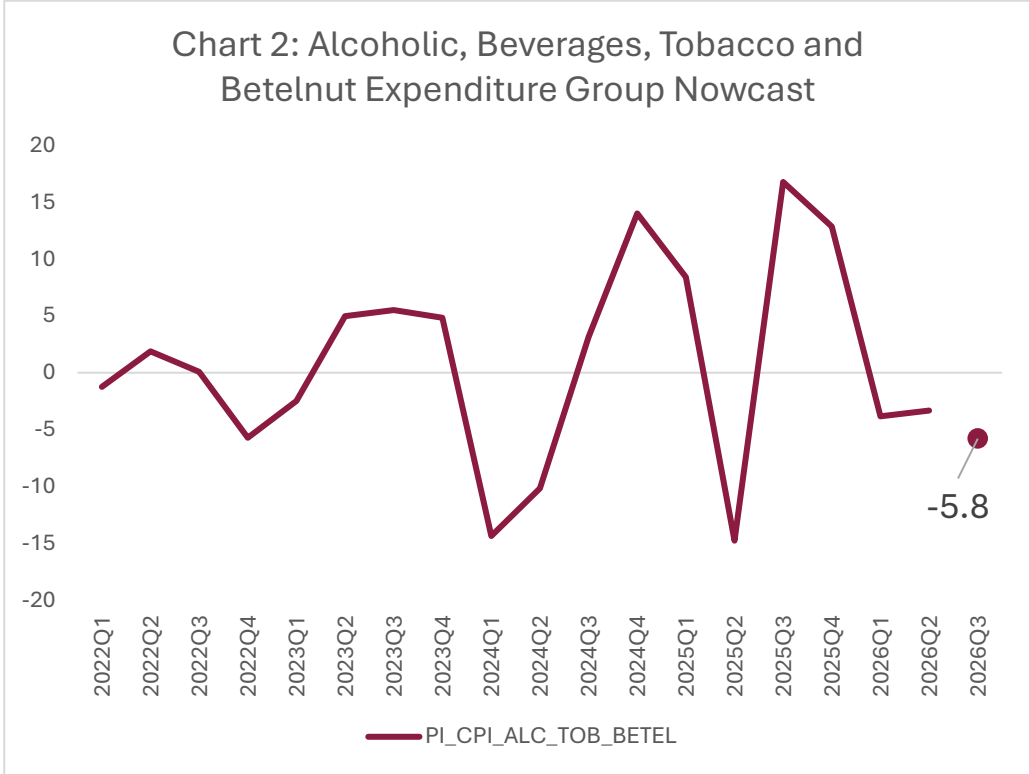
Quarterly Nowcast =1.0%

Annual Nowcast = 5.6%

Method 2: Nowcasting Headline Inflation by Expenditure Groups

Betelnut, Alcoholic, Beverages and Tobacco Expenditure Group





Method 2: Nowcasting Headline Inflation by Expenditure Groups

Food Expenditure Group

Chart 3: High Frequency Indicators for Food Expenditure CPI

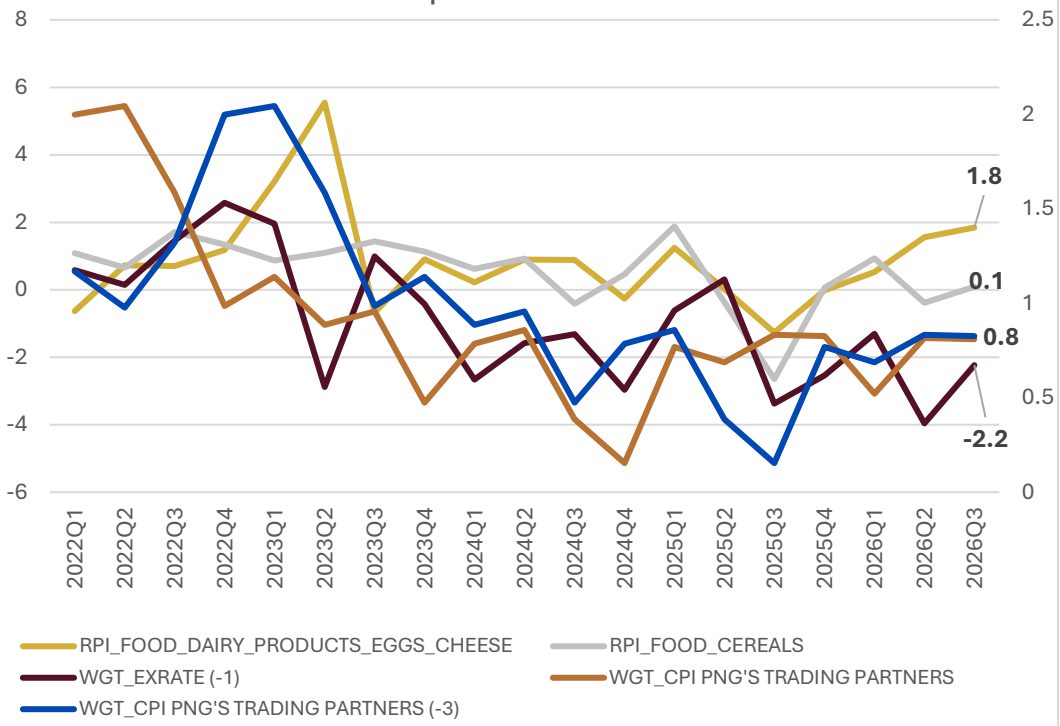
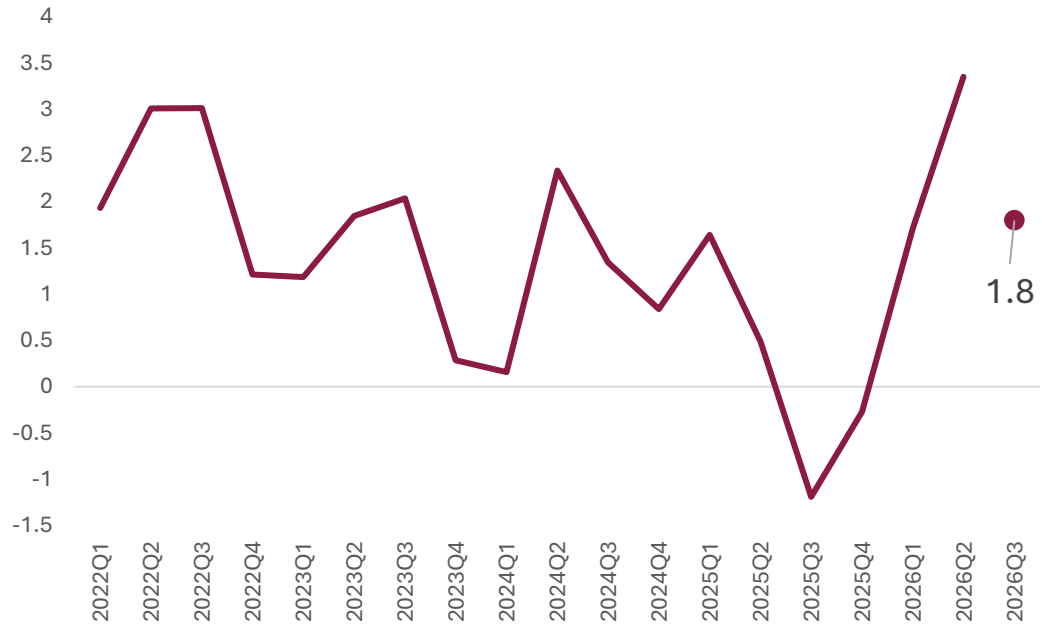


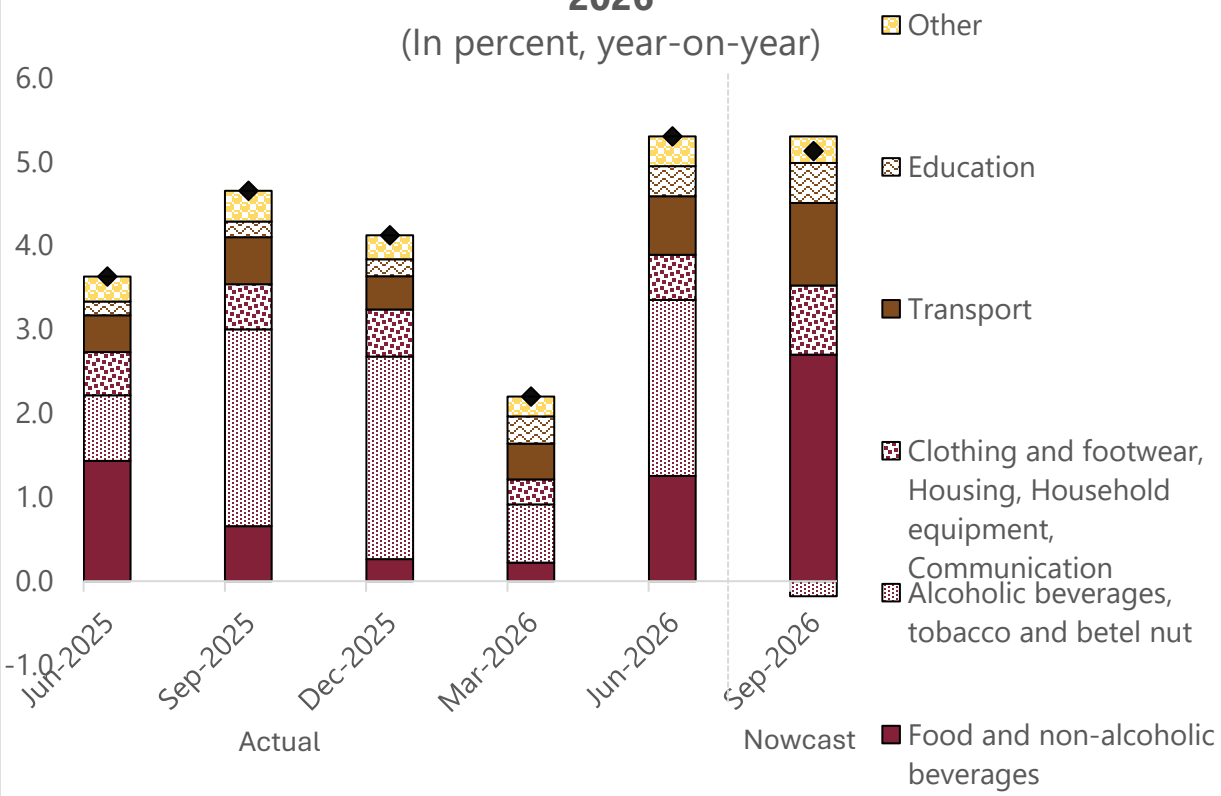
Chart 4: Food Expenditure Group Nowcast



Method 2: Contributions by Expenditure Group to Overall Headline Inflation Nowcast

Chart 5: Contribution by Expenditure Groups: Nowcast September 2026

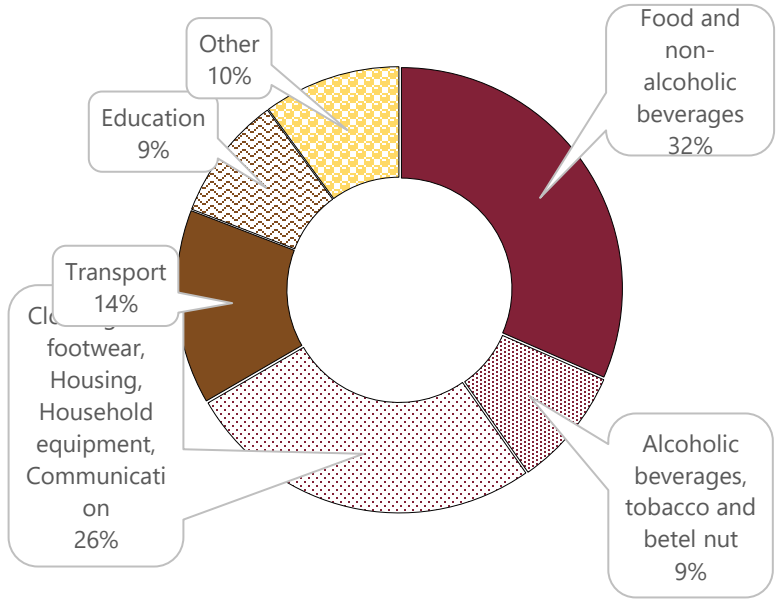
(In percent, year-on-year)



Source: National Statiscal Office and Bank of PNG

Consumer Price Inflation: Weights

(In percent)



Source: National Statiscal Office and Bank of PNG

5. Challenges and Way forward

Challenges

1. Data issues- lack of centralized database, delay in data releases, etc.
2. Specialized nowcasting staff- Competing tasks make it challenging for key staff to focus and specialize in managing the nowcasting/FPAS tools and processes.

Way forward

- Establishment of Statistics department and modelling and forecasting unit in the MEP Group
- Improvements to data collection and data management systems
- Further development and refinement of our nowcasting tools and models
- Documentation of Nowcasting/FPAS process- (work in progress)

6. Conclusion

Thank You!